

DECODED EXPLAINED

4 July, 2026

Building water security in a rapidly drying India

Indian cities – from Bengaluru to Mussoorie (Uttarakhand) – are reeling under severe water stress. June saw a monsoon rainfall deficit of over 40%. Delhi is facing a severe shortage, and water supply has plummeted to about 70% of the total demand of 1,250 million gallons per day. Research by the Council on Energy, Environment and Water (CEEW) further finds that 11 out of 15 major river basins in India are experiencing water stress, with annual water availability below 1,700 m³ per person. Several of them, including the Krishna, Cauvery, Mahi, and Tapi basins, are below the water scarcity threshold of 1,000 m³ per person.

A recent report by the United Nations University Institute for Water, Environment and Health (UNU-INWEH) warns of global water bankruptcy. Several river basins are polluted and closed, with no outflow to the sea. Aquifers are being depleted beyond safe limits. Nearly three-quarters of the global population lives in water-insecure countries, with four billion people facing severe water scarcity for at least one month a year.

Water infrastructure gaps

India has an uneven distribution of water resources. With 4% of the world's water resources supporting 18% of the population, historically, the country has lived and grown with water uncertainty. While schemes such as the Jal Jeevan Mission and Pradhan Mantri Krishi Sinchayee Yojana have helped develop water supply infrastructure, concerns remain regarding the poor upkeep of existing infrastructure, inadequate wastewater treatment facilities, substantial conveyance losses, high levels of water pollution, and low cost recovery.

There are four interrelated actions that can help India navigate the 'state of bankruptcy' to become water secure.

First, invest in climate-proofing water systems



Nitin Bassi

Fellow at the Council on Energy, Environment and Water (CEEW)

The monsoon cannot fix India's water problem; policy could

through granular climate risk assessments of water infrastructure and services. The CEEW's analysis of India's urban flood risk management shows that such assessments help prioritise investments in high to very high-risk locations, including coastal and low-lying areas such as Thane (Maharashtra) and Navsari (Gujarat), or areas with critical infrastructure such as schools, hospitals, and electricity grids, as seen in Ahmedabad. These assessments are especially important for cash-strapped urban local bodies and panchayats. Cities should leverage existing mechanisms like the Urban Challenge Fund (UCF) to finance them—for instance, Visakhapatnam (Andhra Pradesh) recently secured ₹1,501 crore under the UCF for water supply and drainage, part of which could support water risk assessments.

Second, enable water reuse for non-potable purposes in urban areas. Shifting from a linear to a circular approach to water use is essential to reduce pressure on freshwater resources. Used water can be treated and reused for activities such as car washing, landscaping, and cooling data centres. City-level reuse planning is key to achieving this. The Thane Municipal Corporation is adopting a scientific approach to scale up treated wastewater reuse, including for construction, to reduce its annual freshwater deficit of 53 million litres per day and generate revenue. According to CEEW analysis, the economic opportunity from the sale of treated wastewater could reach ₹3 lakh crore and generate 1,00,000 additional jobs by 2047.

Third, scale up micro-irrigation systems, including drip and sprinkler technologies that deliver water to crops more efficiently than flood irrigation. Expanding these systems beyond the current 20% coverage of India's potential 72 million hectares of irrigated area requires large-scale interventions. First, the CEEW's Gujarat research shows that micro-irrigation

subsidies should be redesigned for small and marginal farmers by using 0.4 hectares instead of one hectare as the base unit. Second, farmers need support to shift to low-water, higher-value crops such as horticulture and oilseeds to recover costs, as seen in Andhra Pradesh's Rayalaseema region. Third, low-cost insurance with faster claims, including through a strengthened Pradhan Mantri Fasal Bima Yojana, can help smallholders manage climate and crop risks.

Close water data gaps

Finally, generate data at the river basin level to improve decision-making. India has good data on water availability, but limited data on withdrawals, losses, and consumption at the basin scale. This makes it difficult to assess actual water use, gauge efficiency gains, or allocate water judiciously – often leading one sector to gain at the expense of others, while the lack of data encourages “free riding”, with users extracting as much as the source allows. Artificial Intelligence-based monitoring of water conveyance infrastructure can help detect and measure losses and also generate data. Cities such as Delhi and Bhubaneswar (Odisha) are installing smart bulk water meters to identify and reduce physical losses during distribution. Equally important is analysing end-user consumption by scaling up smart metering and using advanced water accounting tools. India's rollout of over 4.93 crore smart electricity meters can serve as a model for the water sector.

Water is an economic resource – it powers lives, livelihoods, and ecosystems. Reversing water bankruptcy will be the foundation of social welfare. Political will, transparent governance, and societal trust are the levers. Pull them, and India can still turn the tide toward a water-secure future.

Building water security in a rapidly drying India

- ❑ 11 out of India's 15 major river basins are water-stressed, and several have crossed the water-scarcity threshold.
- ❑ Climate change, declining groundwater levels, pollution, and uneven water distribution are worsening the crisis.
- ❑ India has only **4% of the world's freshwater resources** but supports **18% of the global population**.
- ❑ UNU- INWEH- warns of **Global water bankruptcy**.

Global Water Bankruptcy- **Global Water Bankruptcy** is a term used to describe a situation where **water demand exceeds the sustainable supply of freshwater resources**, causing depletion of rivers, lakes, aquifers, and ecosystems.

Causes

- **Over-extraction of groundwater** for agriculture and industries.
- **Population growth** increasing water demand.
- **Climate change** causing droughts and irregular rainfall.
- **Pollution** reducing usable freshwater.
- **Inefficient irrigation and water management**.

Building water security in a rapidly drying India

□According to international standards, water stress occurs when annual per capita water availability falls below 1,700 cubic metres.

□Water Scarcity is reached when annual per capita water availability falls below 1,000 cubic metres.

□Major Challenges - Water Infrastructure Gaps

Despite schemes like: **Jal Jeevan Mission (JJM)** and **Pradhan Mantri Krishi Sinchayee Yojana (PMKSY)**

□Problems remain:

□Poor maintenance of infrastructure-leakages, siltation, seepage losses

□Inadequate waste water treatment

□High conveyance losses during distribution

□Water pollution

□Slow recovery of water service costs- construction, electricity for pumping, treatment and purification, maintenance and repairs

Building water security in a rapidly drying India

Four Key Solutions Suggested

1. Climate-Proof Water Systems

- Assess climate risks to water infrastructure.
- Identify vulnerable locations and critical infrastructure.
- Prioritize investments in high-risk areas.
- Use financing mechanisms such as the **Urban Challenge Fund (UCF)**.

Example: Visakhapatnam secured funds under UCF for water supply and drainage projects.

2. Promote Water Reuse and Circular Water Economy

- Treat and reuse wastewater for non-drinking purposes such as: Construction, Landscaping, Car washing
Cooling systems etc.
- Reduce dependence on freshwater resources.
- Generate revenue and employment from treated wastewater.

Example: Thane Municipal Corporation is scaling up wastewater reuse to reduce freshwater shortages.

3. Expand Micro-Irrigation in Agriculture

Promote:

Drip irrigation and Sprinkler irrigation

- Improve water-use efficiency compared to flood irrigation.
- Redesign subsidies to better support small and marginal farmers.
- Encourage a shift towards: Horticulture, Oilseeds, Other low-water, high-value crops
- Strengthen crop insurance under **Pradhan Mantri Fasal Bima Yojana (PMFBY)**.

4. Close Water Data Gaps

- Generate reliable data at the river-basin level.
- Improve monitoring of: Water withdrawals, Consumption, Distribution losses
- Use: Smart water meters, Artificial Intelligence (AI), Advanced water accounting systems
- Better data can prevent over-extraction and improve allocation efficiency.

Examples: Delhi and Bhubaneswar are deploying smart bulk water meters.

Conclusion

The article argues that **monsoon rains alone cannot solve India's water crisis**. India needs:

- Climate-resilient water infrastructure,
- Wastewater reuse,
- Efficient irrigation,
- Better water data and governance.
- With strong political commitment, transparent governance, and public participation, India can move towards a **water-secure future**.

Building water security in a rapidly drying India

The **Urban Challenge Fund (UCF)** is a scheme announced in the **Union Budget 2025–26** to support the transformation of Indian cities into engines of economic growth through better planning, infrastructure, and governance.

Objectives

The fund aims to:

- Promote planned urbanization**
- Support **city redevelopment projects**.
- Improve **urban infrastructure** such as roads, water supply, sanitation, and public transport.
- Encourage **transit-oriented development (TOD)** around metro and transport corridors.
- Strengthen cities as centers of investment, innovation, and employment.

Key Features

- An allocation of **₹1 lakh crore** has been proposed.
- The fund seeks to attract both **public and private investment**.
- States and Urban Local Bodies (ULBs) are expected to contribute their share and undertake reforms to access the fund.

Building water security in a rapidly drying India

❑ **Pradhan Mantri Krishi Sinchai Yojana (PMKSY)** is a **centrally sponsored scheme** launched in **2015** with the objective of expanding irrigation coverage and improving water use efficiency through "**Har Khet Ko Pani**" and "**Per Drop More Crop**". It is run by the **Ministries of Agriculture, Jal Shakti and Rural Development**.

Four main components of PMKSY:

- ❑ **Accelerated Irrigation Benefits Programme (AIBP):** To expedite completion of large and medium irrigation projects.
- ❑ **Har Khet Ko Pani (HKKP):** Undertaking Command Area Development (CAD) by creating new water sources and expanding distribution networks.
- ❑ **Per Drop More Crop (PDMC):** Encouraging the use of drip/sprinkler irrigation systems at the farm level, bringing efficiency in the use of fertilizers.
- ❑ **Watershed Development:** Conservation of natural resources (soil, vegetation and water), groundwater recharge and promotion of agro-forestry.

Building water security in a rapidly drying India

❑ **Jal Jeevan Mission** - Launched on **15th August 2019**, the **Jal Jeevan Mission (JJM)** aims to provide **tap water to every rural household by 2024 (extended till 2028)**, targeting **55 liters per person per day**. It is a **Centrally Sponsored Scheme**, implemented by the **Ministry of Jal Shakti**.

❑ **Pradhan Mantri Fasal Bhima Yojana** - The PMFBY is a **Central Sector scheme** launched in **2016**, designed to offer **financial protection to farmers against crop losses caused by natural disasters, pests, or diseases**. The scheme is voluntary.

Govt's plan to revive struggling tiger reserves

Nikhil Ghanekar
New Delhi, July 3

AN EVENT marking the 18th anniversary of tiger reintroductions at the Sariska Tiger Reserve in Alwar, Rajasthan, last month saw the Centre release two new assessments. The first was a road map to manage tigers going forward and the second reflected on the lessons learnt from 12 reintroduction initiatives in India.

It has been argued that beyond simply counting over-all numbers, the focus should be on national parks and reserves with a low number of tigers. To aid the recovery of big cats, parks with poor habitats and prey availability have also been included. This, the Centre said, will also aid the dispersal of tigers from places which have reached peak carrying capacity towards newer habitats.

Numbers rise, but only in pockets

India's tiger population has risen steadily. From 1,411 in 2006, there are now 3,682 tigers in 2022 across 58 tiger reserves, spread across 85,000 sq km. The numbers, however, do not reveal the full story.

Of the 58, 10-12 reserves alone account for about 36% of the population. There are 12 tiger reserves with less than three tigers inside and of them, three — Kawal, Kamlang and Dampa — have zero tigers.

As the density of tigers increases, they disperse to buffer areas, territorial divisions and mixed-use land. This often puts them in conflict with humans while increasing their dependency on livestock on these fringes, besides elevating the risk of poaching, and mortality while crossing railways, roads and canals.

A MIXED RECORD

- Sariska, which had previously lost all its tigers, was the first to undertake reintroduction in 2006. Panna in MP followed suit in 2009 — 10 more reserves have done so since then.

- The Centre acknowledged failure in Odisha's Satkosia owing to rejection by local communities, besides slow and lethargic progress in Rajasthan's Mukundara Hills.

Parks with low tigers, however, may have intact forests but are also low on prey. Neither situation is ideal.

The unevenness between "source populations" (where habitat, prey and tiger numbers are high) and "sink populations" (where there are no breeding tigers or connectivity with healthier forest) poses a challenge for long-term conservation.

The Centre's latest assessment calls for consolidating "source populations" in 13 tiger reserves such as Corbett, Bandipur and Kaziranga. It has suggested the need for priority interventions in at least 25 reserves, including reintroductions where less than five tigers remain.

Need for 'recipient sites'

Tiger population growth has sustained at 6% annually. Uneven growth, however, means that regions with high density are facing the pressure of managing dispersing tigers outside forests, human-wildlife conflict and poaching pressures. This comes even as vast forests are bereft of tigers. This situation arises due to the fragmentation of forested areas, tigers being discouraged to move across long distances, poor prey in sink areas forcing tigers to vacate them and external pressure from human activities. A well-connected landscape of forests across tiger reserves, territorial forests, mixed-use areas is key for establishing a metapopulation, helping in genetic exchange and reducing extinction risk in the long-term.

The assessment notes that long-term recovery of tiger populations depends on the interplay of prey abundance, habitat quality and colonisation by tigers themselves.

Government's Plan to Revive struggling Tiger Reserves

□ On the **18th anniversary** of tiger reintroductions at Sariska Tiger Reserve (Rajasthan), the Centre released:

- A **roadmap for future tiger management**.
- An assessment of **12 tiger reintroduction initiatives** in India.

1. India's Tiger Population: Success with Uneven Distribution

- Tiger numbers increased from **1,411 (2006)** to **3,682 (2022)**.
- These tigers are spread across **58 tiger reserves covering about 85,000 sq. km.**
- However, growth is highly uneven.

2. Concentration of Tigers

- **12 reserves account for about 36%** of India's tiger population.
- **12 tiger reserves have fewer than 3 tigers.**
- **Three reserves have zero tigers:**

Kawal Tiger Reserve

Kamlang Tiger Reserve

Dampa Tiger Reserve

Takeaway: Tiger recovery is occurring only in a few "successful" reserves, while many reserves remain underpopulated.

Government's Plan to Revive struggling Tiger Reserves

- ▣ **Kawal Tiger Reserve- Telangana-** Declared a Tiger Reserve in 2012; located along the Godavari river basin; important tiger corridor connecting central and southern India.
- ▣ **Kamlang Tiger Reserve- Arunachal Pradesh- India's 50th Tiger Reserve;** notified in 2017; located in Lohit district near **Namdapha Tiger Reserve.**
- ▣ **Dampa Tiger Reserve- Mizoram-** Largest wildlife sanctuary/tiger reserve in Mizoram; declared Tiger Reserve in 1994 under Project Tiger.

- ▣ Why Low Tiger Numbers are a Problem
 - **High-density reserves-**When tiger populations increase beyond a reserve's carrying capacity, tigers move into: Buffer areas, Territorial forests, Mixed-use landscapes. This leads to: Human–tiger conflict, Livestock depredation, Higher poaching risk, Mortality from roads, railways, and canals
 - **Low-density reserves-**Often possess good forest habitat. But suffer from: Low prey base and Lack of breeding populations.
 - **Neither situation is ideal.**

Government's Plan to Revive struggling Tiger Reserves

Source and Sink Populations

Source Population : Areas where

- Habitat Quality is high
- Prey is abundant
- Tiger number are healthy
- Surplus tigers disperse outward
 - Examples: Corbett Tiger Reserve, Bandipur Tiger Reserve, Kaziranga Tiger Reserve

Sink Population : Areas where:

- Tiger numbers remain low.
- Breeding populations are absent or weak.
- Connectivity with healthier forests is poor
- **Conservation challenge:** Connect source and sink populations through wildlife corridors.

4. Government's Strategy

1. Focus on Weak Reserves

Instead of only counting tiger numbers, the Centre proposes: Reviving reserves with low tiger populations, Improving habitat quality, Increasing prey availability, Facilitating dispersal of tigers from overcrowded reserves.

2. Priority Intervention

- Consolidate source populations in **13 tiger reserves**.
- Undertake focused interventions in **at least 25 reserves**.
- Consider reintroductions where tiger numbers are **below five**.

5. Importance of Landscape Connectivity

The assessment emphasizes:

Creating **well-connected forest landscapes** across: Tiger reserves, Territorial forests, Mixed-use areas

Benefits: Formation of a **metapopulation** (interconnected tiger populations), Genetic exchange, Reduced inbreeding, Lower extinction risk, Natural colonization of new habitats.

Metapopulation

A group of geographically separated populations of the same species connected through migration and dispersal

Government's Plan to Revive struggling Tiger Reserves

6. Factors Needed for Long-Term Recovery

The report highlights that tiger recovery depends on: **Adequate prey abundance** (deer, wild boar, gaur, etc.), **High-quality habitat**, **Habitat connectivity/corridors**, **Natural colonization by dispersing tigers**.

Reduction in human pressures and fragmentation.

7. Lessons from Reintroduction Efforts

Success : Sariska Tiger Reserve became the first reserve where tigers were successfully reintroduced after local extinction (2008).

Mixed Results

Panna Tiger Reserve followed in 2009 and became a notable success.

Challenges

Failure in Odisha's Satkosia Tiger Reserve due to local community objections.

Slow progress in Mukundara Hills Tiger Reserve, Rajasthan

Burning bright: tiger population quadruples at Assam's Nameri Tiger Reserve in 3 years

Rahul Karmakar

GUWAHATI

The population of tigers at the 344 square km Nameri Tiger Reserve in north-central Assam's Sonitpur district has quadrupled in three years, the State's Forest and Environment Minister Jayanta Mallabaruah said on Friday.

One of four tiger reserves in Assam, Nameri shares a boundary with the 1,276.95 sq. km Pakke Tiger Reserve in Arunachal Pradesh. The other three tiger reserves are Kaziranga, Manas, and Orang, the first two being UNESCO World Heritage Sites.

Mr. Mallabaruah said the Wildlife Institute of India (WII) validated the State Forest Department's 2025-end estimation of 12



One of four tiger reserves in Assam, Nameri shares a boundary with Pakke Tiger Reserve in Arunachal Pradesh. FILE PHOTO

tigers at Nameri. This was a four-fold increase from three striped cats recorded during the All India Tiger Estimation (AITE) of 2022.

He attributed the "extraordinary results" to the forest conservation efforts under the guidance of Chief Minister Himanta Biswa Sarma.

"From just 3 tigers in 2022 to 12 by the end of 2025, Nameri Tiger Reserve has scripted a remarkable conservation success," he said, also highlighting the return of tigers to the Sonai Rupai Wildlife Sanctuary after more than two decades.

The tiger went locally

extinct at Sonai Rupai, Nameri's satellite core, in the early 2000s. "Scientific monitoring and long-term habitat protection are strengthening tiger conservation in the Nameri-Sonai Rupai landscape," Mr. Mallabaruah said.

Kaziranga status

Better known for its one-horned rhinos, Kaziranga has one of the highest tiger densities – 13.44 per 100 sq. km – in India.

The 2022 AITE reported that Kaziranga had 104 tigers, one more than the number recorded in the 2014 AITE. However, 148 tigers were recorded in the 'Status of Tigers in Kaziranga Tiger Reserve', a report released by State Forest Department in 2024 stated.

Burning bright: tiger population quadruples at Assam's Nameri Tiger Reserve in 3 years

Numbers increased from **3 tigers in 2022** (All India Tiger Estimation) to **12 tigers by the end of 2025**.

This represents a significant conservation success in a reserve that previously had very low tiger numbers.

Location and Connectivity

Nameri Tiger Reserve : Located in **Sonitpur district, Assam**.

Area: **344 sq. km**.

Shares a boundary with **Pakke Tiger Reserve**.

Importance

Connectivity with Pakke creates a larger tiger landscape.

Facilitates: Tiger dispersal, Genetic exchange, Healthy breeding populations,

Tiger Reserves in Assam- Assam has **four tiger reserves**:

Kaziranga Tiger Reserve, Manas Tiger Reserve, Orang Tiger Reserve, Nameri Tiger Reserve

Tiger Reserves India



Burning bright: tiger population quadruples at Assam's Nameri Tiger Reserve in 3 years

UNESCO World Heritage Sites -Kaziranga and Manas

Validation of Tiger Numbers

The estimate of **12 tigers in 2025** was validated by:

Wildlife Institute of India (WII).

All India Tiger Estimation (AITE)

Conducted periodically by National Tiger Conservation Authority and Wildlife Institute of India to estimate tiger populations across India.

The 2022 AITE recorded only **3 tigers** in Nameri. In 2025, it increased to 12.

Kaziranga had 102 tigers in 2022 and increased to 103 in 2025.

Reasons for Success

According to Assam's Forest Department, the increase resulted from:

Scientific monitoring.

Long-term habitat protection.

Strong forest conservation efforts.

Improved tiger conservation in the Nameri–Sonai Rupai landscape.

Burning bright: tiger population quadruples at Assam's Nameri Tiger Reserve in 3 years

❑ **Sonai Rupai Wildlife Sanctuary** -Serves as a satellite core area linked to Nameri.

Tigers had become **locally extinct** there in the early 2000s.

Tigers have now returned after more than two decades.

Significance

Demonstrates successful recolonization of habitat.

Shows the importance of connected landscapes.

Kaziranga's Tiger Status- High Tiger Density- Kaziranga has one of the **highest tiger densities** in India:

13.44 tigers per 100 sq. km.

State Forest Department Report (2024): 148 tigers recorded.

Interpretation

Tiger numbers continue to rise in Kaziranga, making it one of India's strongest tiger source populations.

Link with National Tiger Conservation Strategy

This article supports the broader national strategy of:

Strengthening **source populations**, Improving **habitat connectivity**, Facilitating **natural dispersal**.

Reviving low-tiger reserves without relying solely on translocation.

Matcha tea, now from India: Assam estate sells first batch

Sukrita Baruah
Guwahati, July 3

FEATURING IN lattes and desserts across trendy cafes, matcha tea has found growing demand in India. And on Friday, the first batch of Indian-manufactured matcha tea, produced at a tea estate in Assam, was sold at the Guwahati Tea Auction Centre (GTAC).

The five-kg batch of matcha produced at the Chota Tingrai Tea Estate in Assam's Tinsukia district was sold for Rs 3,000 at the GTAC to a Guwahati-based buyer. Currently, the matcha sold in Indian markets is imported from Japan, China and Vietnam.

Mritunjay Jalan, director of Chota Tingrai Tea Estate, said that they had set up a Japanese tea manufacturing facility on their estate in 2016 and had been working with experts from the country since then to produce green tea, but decided to venture specifically into matcha this year after observing a jump in demand.

"We had begun 10 years ago when we realised that there was a big demand for green tea starting in India, but that the Indian market did not have authentic green tea... But the demand for matcha had not picked up at the time and it is a very specialised product. You cannot mass-produce it, and we didn't want to jump ahead because the market was not ready, and investments are high. But last year, we saw such a huge demand for matcha globally, which has also reached India, and we thought now is the

right time to start," he said.

Assam Chief Minister Himanta Biswa Sarma highlighted this move after Friday's sale.

"Furthering our glorious tea legacy, Assam becomes the first state in India to commercially produce matcha tea at the Chota Tingrai Tea Estate in Tinsukia. This diversification of the viral drink will help strengthen Brand Assam Tea in the global markets. This has been made possible due to the strong India-Japan ties and our collaboration in various fields, and has fetched a handsome price in the Guwahati Tea Auction Centre," he said.

The market for Assam tea has been grappling with several issues for a long time, not least of which, multiple stakeholders say, is that production outweighs the demand. Different stakeholders often hold up the global boom in demand for matcha as a contrast to what they say is a largely stagnant demand for Assam tea, with the challenges in the industry pushing several estates into diversifying into sectors such as tea tourism.

The investment in producing matcha is sizable, Jalan says, and production capacity is roughly 1/10 of that of a black tea factory. For now, he says, the target market for them is domestic.

"Tea estates in Assam will have to adapt and diversify because the black tea market has stagnated, with the result that when there is good production, prices fall... We have diversified into tea tourism, but are also diversifying within tea, trying to provide the new age products that are in demand," he said.

Matcha tea, now from India: Assam estate sells first batch

Assam has become the **first Indian state to commercially produce matcha tea**.

The first batch was produced at **Chota Tingrai Tea Estate**.

The first batch was sold through the **Guwahati Tea Auction Centre (GTAC)**.

First Commercial Sale

A **5 kg batch of matcha tea** was auctioned.

It fetched **₹3,000 per kg**.

Significance

Marks the beginning of indigenous matcha production in India.

Reduces dependence on imports.

Matcha Market in India

Matcha is increasingly popular in: Lattes, Desserts, Cafés, Health and wellness products

Until now, most matcha sold in India was imported from: Japan, China, Vietnam

Matcha tea, now from India: Assam estate sells first batch

Key Challenge

- Matcha is a specialized product.
- Cannot be mass-produced easily.
- Requires significant investment and expertise.

Importance for Assam Tea Industry

Existing Problem

The Assam tea industry faces:

- Stagnant demand for conventional black tea.
- Oversupply relative to demand.
- Falling prices during periods of high production.

Economic Significance

Brand Assam Tea

According to the Assam government, Commercial matcha production will-

Strengthen **Brand Assam Tea** globally, Open new premium tea markets, Increase value addition in the tea sector.

Matcha tea, now from India: Assam estate sells first batch

Diversification Strategy

Helps tea estates move beyond dependence on black tea.

Provides access to high-value niche markets.

India–Japan Cooperation

The initiative was supported by collaboration with Japanese experts.

Highlighted as an outcome of strengthening India–Japan relations, technology transfer and agricultural value addition.

Production Constraints

-Matcha manufacturing requires specialized processing.

-Investment costs are high.

-Production capacity is only about **one-tenth that of a black tea factory**.

‘No revival of Indus treaty if Pak. does not end terror’

Kallol Bhattacharjee

NEW DELHI

India on Friday said the Indus Waters Treaty (IWT) will remain in “abeyance” till Pakistan “credibly and irrevocably” stops supporting cross-border terrorism. Official spokesperson of the External Affairs Ministry Randhir Jaiswal made the remark in the context of a recent international conference held in Pakistan where it called for the restoration of the treaty. The country is dealing with a severe heatwave. “The Indus Waters Treaty of 1960 will be held in abeyance with immediate effect, until Pakistan credibly and irrevocably abjures its support for cross border terrorism,” said Mr. Jaiswal at a media briefing.

Earlier this week, Pakistan organised a conference on the Indus and water sharing where the country’s Foreign Minister and Deputy Prime Minister Ishaq Dar referred to the IWT



Point of contention: India’s Cabinet Committee on Security had put the IWT in abeyance after the Pahalgam attack. PTI

as vital for “regional peace, stability and cooperation”.

The Cabinet Committee on Security had put the treaty in abeyance a day after the Pahalgam terror attack on April 22, 2025, killed 26 civilians and injured dozens. The attack led to India’s strikes on terror targets inside Pakistan under Operation Sindoor.

“The IWT was concluded in the spirit of goodwill and friendship as specified in the preamble of the treaty. However, Pakistan has held these principles in abeyance by its promotion

of cross-border terrorism for decades,” Mr. Jaiswal had said on May 13, 2025.

On May 22 last year, Mr. Jaiswal had said that India was willing to discuss with Pakistan the modality to get custody of “noted terrorists whose list was provided to Pakistan some years ago”. Citing PM Narendra Modi’s remarks, he said, “Water and blood cannot flow together”. India has intensified the rhetoric on Pakistan recently, especially against the backdrop of protests in Pakistan Occupied Kashmir.

No revival of Indus treaty if Pak. does not end terror

The **Indus Waters Treaty** was signed on **19 September 1960** between **India** (Nehru) and **Pakistan** (Ayub Khan), with the **World Bank** acting as a mediator.

1. Division of Rivers

The six rivers of the Indus system were divided into two groups:

Eastern Rivers (allocated to India)-Ravi River , Beas River ,Sutlej River

India has **exclusive rights** over the waters of these rivers.

Western Rivers (allocated primarily to Pakistan)- Indus River , Jhelum River ,Chenab River

Pakistan receives the bulk of the waters from these rivers.

2. Limited Rights for India on Western Rivers

India can Use water for domestic purposes, Irrigate a limited area, Generate **run-of-the-river hydropower** projects, Navigate and fish in these rivers ,India cannot substantially alter or stop the natural flow of water.

3. Permanent Indus Commission

A **Permanent Indus Commission (PIC)** was established.

Each country appoints one commissioner.

The commission exchanges data and resolves routine issues.

4. Dispute Resolution Mechanism

The treaty provides a three-level mechanism: Bilateral discussions through the PIC, Neutral Expert for technical disputes. ,Court of Arbitration for legal disputes.

The Indus Waters Treaty (IWT)

- The distribution of waters of the Indus and its tributaries between India and Pakistan is governed by the Indus Water Treaty (IWT).
- Was signed on Sept 19, 1960, between India, Pakistan and a representative of World Bank after eight years of negotiations.
- Partition of India cut across the Indus river basin, which has the Indus river, plus five of its main tributaries.

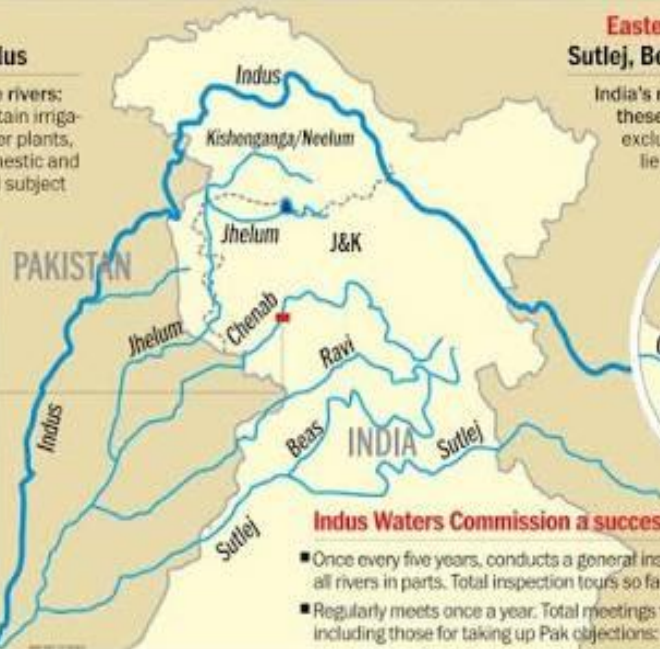
Western rivers

Chenab, Jhelum, Indus

India's rights over these rivers: Limited — can set up certain irrigation, run-of-the-river power plants, very limited storage, domestic and non-consumptive use, all subject to conditions



Baglihar dam on Chenab



Eastern rivers

Sutlej, Beas, Ravi

India's rights over these rivers: All exclusive rights lie with India.



Indus Waters Commission a success story

- Once every five years, conducts a general inspection of all rivers in parts. Total inspection tours so far: Over 100
- Regularly meets once a year. Total meetings thus far, including those for taking up Pak objections: Over 100

No revival of Indus treaty if Pak. does not end terror

Suspension of IWT

❑ In 2023, India issued its first notice under the IWT, requesting modifications to the Treaty, focusing on the [Kishenganga](#) and [Ratle](#) hydroelectric projects. Despite these projects being classified as **"run-of-the-river"** and intended to **generate electricity** without obstructing the **natural river flow**, Pakistan raised concerns, claiming they violate the IWT's stipulations.

❑ India's suspension marks the first suspension of the treaty since its inception, signaling a **shift in water diplomacy linked to cross-border terrorism**. [Article 62 of the Vienna Convention](#) allows a country to withdraw from or repudiate a treaty if there is a fundamental change in circumstances that makes the continuation of the treaty untenable.

- ❑ India can stop sharing river data
- ❑ No design or operational restrictions
- ❑ Can create storage on the western rivers
- ❑ Can restrict joint inspection

No revival of Indus treaty if Pak. does not end terror

- ❑ **Beas**- Beas Project, Pong Dam, Pandoh Dam
- ❑ **Chenab**- Salal Project, Dhulhasti Dam, Baglihar Dam, Trimmu Barrage, Ratle Hydro Project
- ❑ **Jhelum** - Mangla Dam, Rasul Barrage
- ❑ **Ravi**- Thein (Ranjit Sagar) Dam
- ❑ **Sutlej**- Bhakra Nangal Project, Harike, Sirhind, Gobind Ballabh Sagar, Karcham Wangtoo Hydroelectric Plant, Neptha Jhakhari Dam

No revival of Indus treaty if Pak. does not end terror

India's Position on the Indus Waters Treaty (IWT)

India has stated that the **Indus Waters Treaty (1960)** will remain in "**abeyance**" (temporarily suspended).

The treaty will not be restored until Pakistan "**credibly and irrevocably**" ends its support for **cross-border terrorism**.

This statement was made by **Randhir Jaiswal**, spokesperson of the Ministry of External Affairs (MEA).

Reason for Keeping the Treaty in Abeyance

The decision is linked to India's concerns about **cross-border terrorism originating from Pakistan**.

India argues that the treaty was signed in a spirit of **goodwill, friendship, peace, stability, and cooperation**, but Pakistan has violated these principles by supporting terrorism.

The **Cabinet Committee on Security (CCS)** placed the treaty in abeyance after the **Pahalgam terror attack**.

No revival of Indus treaty if Pak. does not end terror

India's Response

Following the attack, India carried out strikes against terror targets in Pakistan under **Operation Sindoor**.

India linked the continuation of the treaty to Pakistan's actions against terrorism.

Pakistan's Demand

Pakistan recently hosted an international conference on the Indus River and water sharing. Pakistani leaders called for the **restoration of the Indus Waters Treaty**, emphasizing its importance for: Regional peace, Stability, Cooperation

India's Rejection of Pakistan's Demand

India reiterated that there will be **no revival of the treaty** unless Pakistan completely abandons support for terrorism.

India maintains that terrorism and normal bilateral cooperation cannot coexist.

Antibiotics to creams: The perils of combination meds



EXPERT EXPLAINS

KAMINI WALLA

SENIOR SCIENTIST AT INDIAN COUNCIL OF MEDICAL RESEARCH

THE GOVERNMENT has banned 16 fixed-dose combination (FDC) drugs, including certain antibiotic combinations and a range of dermatological products containing aloe vera and other herbal ingredients, because their supposedly amplified benefits lack scientific justification.

FDCs contain two or more active ingredients in a single formulation and are commonly prescribed for conditions ranging from infections to pain and skin ailments. While some combinations are evidence-based and improve patient outcomes, others are considered "irrational" because there is little or no scientific evidence that the ingredients work better together than when used separately. "Public health experts have long warned that irrational combinations can expose patients to unnecessary drugs, increase costs and, in the case of antibiotics, contribute to antimicrobial resistance. The last is a growing public health problem because bacteria, viruses, fungi, and parasites no longer respond to the medicines designed to kill them," Dr Kamini Walla, senior scientist at the Indian Council of Medical Research, tells Rinku Ghosh.

What is an 'irrational' FDC?

An FDC is considered irrational when the ingredients do not have a scientifically established rationale for being combined in a single product. For a combination to be considered rational, each component should contribute meaningfully to the intended therapeutic effect, have compatible pharmacological properties and demonstrate additional clinical benefit compared to using the individual medicines separately. In many cases, there is little to no evidence from clinical trials to support the combination.

One of the banned products combines amoxicillin and serratiopeptidase. Why have experts objected to this?

Serratiopeptidase is a proteolytic enzyme, which breaks down proteins. The evidence supporting its use alongside anti-

• HOW FDCs ARE REGULATED AROUND THE WORLD

U.S.: All FDCs require a new drug application with clinical evidence of superiority or convenience over mono-components.

WHO: WHO explicitly cautions against irrational FDCs. Those in the essential

medicines list are evidence-based.

INDIA (PRE-REFORM): Thousands of FDCs were approved by state licensing authorities without central review, exploiting a regulatory loophole in the Drugs & Cosmetics Act.

INDIA (POST-2016): Around 6,000 FDCs were reviewed by a committee; bans have been initiated.

EUROPEAN UNION: Full scientific review of FDCs; they can be justified only with clinical data.

Some popular combination drugs

NORFLOX TZ (combination of norfloxacin and tinidazole)

- If you just have bacterial diarrhoea, taking tinidazole is pointless.
- If you have amoebic dysentery, norfloxacin provides zero benefit.
- Patients rarely suffer from bacterial and protozoal infections simultaneously. But exposure unnecessarily promotes bacterial resistance.

AUGMENTIN 625 (combination of amoxicillin and clavulanic acid)

- Clavulanic acid blocks enzymes that certain resilient bacteria produce to destroy amoxicillin.
- But if the bacteria causing your infection are not resistant, the clavulanic acid is useless.



COMBINATION CREAMS

(combining a steroid and an anti-fungal)

- While they provide temporary relief from itching and redness, steroids suppress the skin's local immune response.
- This can cause the underlying fungal infection to worsen, spread, or become resistant to treatment.

- DR KAMINI WALLA

biotics is extremely limited. The enzyme is acid-labile, meaning it can be degraded in the stomach before reaching the bloodstream. There is no evidence that shows adequate therapeutic concentrations reach infected tissues. No peer-reviewed randomised controlled trial has shown that adding serratiopeptidase improves bacterial clearance, increases cure rates or reduces the amount of antibiotic required.

Some proponents argue that it may help antibiotics penetrate biofilms or inflamed tissues more effectively. However, this hypothesis is largely based on laboratory studies and has not been convincingly demonstrated in human clinical studies. No major treatment guideline currently recommends serratiopeptidase as an adjunct to antibiotics for managing infections.

What about popular combination drugs in use?

Take the case of the antibiotic Norflox TZ (a combination of norfloxacin and tinidazole). If you have purely bacterial diarrhoea, taking tinidazole is pointless. Conversely, if you have amoebic dysentery, norfloxacin provides zero benefit. Patients rarely suffer from bacterial and protozoal infections simultaneously. But exposure unnecessarily promotes bacterial resistance.

Doctors often caution against the misuse of Augmentin 625 (a combination of

What are FDCs?

FDCs, or fixed-dose combination drugs, contain two or more ingredients in one formulation. They claim to amplify benefits.

Some combinations do improve patient outcomes. But FDCs are called 'irrational' when there is little evidence that the ingredients work better together than when used separately.

amoxicillin and clavulanic acid). The clavulanic acid acts as a barrier to block enzymes that certain resilient bacteria produce to destroy amoxicillin. But if the specific bacteria causing your infection are not resistant, the clavulanic acid is useless.

Could such antibiotic combinations contribute to antimicrobial resistance?

When combinations are marketed as being more effective without sufficient evidence, they may encourage unnecessary or prolonged antibiotic use. This increases antibiotic exposure in the community and creates selective pressure on bacteria, allowing resistant organisms to survive and multiply. From a public health perspective, antibiotic use should be as targeted and evidence-based as possible.

Many aloe vera-based dermatological combinations have also been banned. Why were these products considered irrational despite being sold for years?

Longevity in the market does not establish scientific validity. Many of the banned dermatological combinations contain multiple ingredients such as aloe vera extracts, vitamin E, jojoba oil, olive oil, tea tree oil and other moisturising or herbal components. The key question is whether combining these ingredients provides a

measurable clinical benefit compared with using them individually. In many cases, robust scientific evidence demonstrating superior efficacy is lacking. Take the case of combination creams containing both a steroid and an antifungal. While they provide temporary relief from itching and redness, steroids suppress the skin's local immune response, which can cause the underlying fungal infection to worsen, spread, or become resistant to treatment. Regulatory decisions are based on scientific evidence and risk-benefit assessments, not how long a product has been available commercially.

What risks do patients face when using irrational FDCs?

Patients may be exposed to unnecessary drugs, increasing the possibility of adverse effects, drug interactions and allergic reactions. Fixed combinations can also make it difficult for doctors to adjust the dose of individual ingredients according to a patient's needs. If a doctor wants to increase the dose of one medication, they cannot do so without overloading on the other. Besides, such drugs may mask an underlying complication, reducing precision treatment.

What should patients, doctors and pharmacists do to ensure these products have been banned?

Patients should understand that a medicine containing multiple ingredients is not necessarily more effective than a targeted treatment. In fact, a simpler medicine supported by strong evidence is often the safer and more effective option. Patients currently using any of the banned products should consult their doctor about appropriate alternatives. Stopping an irrational FDC does not mean stopping treatment; safer and evidence-based alternatives are available.

For doctors, the focus should be on de-escalating patients to rational therapies and prescribing medicines supported by clinical evidence. Pharmacists should remain aware of the drug regulator's list of banned FDCs, flag irrational prescriptions where appropriate and educate patients about available alternatives.

Should we team up vitamins and probiotics with antibiotics?

There is no definitive evidence of them being indispensable. While probiotics may be advised by doctors on a case-specific basis, vitamins may not be needed, except by those in a vulnerable group. Antibiotics have limited dosages; you may not need vitamins for a short spell. Consult your doctor.

Antibiotics to creams: The perils of combination meds

What are Fixed-Dose Combination (FDC) Drugs?

FDCs contain **two or more active ingredients in a single formulation**.

They are designed to improve convenience and sometimes enhance treatment outcomes.

However, many combinations are considered **"irrational"** because there is little or no scientific evidence that the ingredients work better together than separately.

Government Action

The Government of India has **banned 16 Fixed-Dose Combination (FDC) drugs**.

These include:

Certain antibiotic combinations

Dermatological creams containing multiple ingredients (such as aloe vera and herbal extracts)

Why were they banned?

Lack of scientific justification.

No proven additional clinical benefit.

Potential risks to patients.

Possibility of contributing to **Antimicrobial Resistance (AMR)**.

Antibiotics to creams: The perils of combination meds

What is an "Irrational" FDC?

- The ingredients do not have a scientifically established reason to be combined.
- Each ingredient does not contribute meaningfully to the intended therapeutic effect.
- The combination does not show better clinical outcomes than individual drugs.
- Evidence supporting superiority over individual medicines is lacking.

Risks of Irrational FDCs

1. Increased Adverse Effects

Patients may be exposed to unnecessary medicines.

Greater risk of: Side effects, Drug interactions ,Allergic reactions

2. Difficulty in Dose Adjustment

Doctors cannot easily adjust one component without affecting the others.

Increasing one drug's dose may overdose another component.

3. Masking Underlying Disease

Symptoms may temporarily improve while the underlying condition worsens.

4. Antimicrobial Resistance (AMR)

Unnecessary antibiotic exposure promotes resistant bacteria.

Resistant microorganisms survive and multiply, making future infections harder to treat.

Antibiotics to creams: The perils of combination meds

Norfloxacin + Tinidazole

Problems:

If the patient has bacterial diarrhoea:

Tinidazole is unnecessary.

If the patient has amoebic dysentery:

Norfloxacin offers little benefit.

Simultaneous bacterial and protozoal infections are uncommon.

Unnecessary exposure can promote antimicrobial resistance.

How FDCs are Regulated Globally

United States

All FDCs require a **new drug application**.

Clinical evidence must demonstrate superiority or convenience over individual components.

World Health Organization (WHO)

Warns against irrational FDCs.

Only evidence-based combinations are included in essential medicines lists.

Antibiotics to creams: The perils of combination meds

European Union

Requires full scientific review.

Approval only if supported by clinical data.

India (Before Reforms)

Thousands of FDCs were approved by state authorities without central review due to regulatory loopholes.

India (After 2016)

Around **6,000 FDCs** were reviewed by an expert committee.

Several irrational combinations were banned.

What Should Patients Do?

More ingredients do **not necessarily mean better treatment**.

A simpler medicine with strong scientific evidence is often safer and more effective.

Manufacturing justice

Use of AI-fabricated judicial 'precedents' amounts to criminal misconduct

But for its serious implications, a judge relying on Artificial Intelligence (AI) hallucinations to arrive at a judicial determination would have been comical. Given the serious miscarriage of justice that such hallucinations can lead to, if used in the judicial processes, the Supreme Court of India has likened it to methyl isocyanate, the poisonous gas that led to the Bhopal gas tragedy in 1984 – “invisible, insidious, and catastrophic by the time anyone notices”. The Court made these observations while setting aside the orders of the National Company Law Tribunal (NCLT) and the National Company Law Appellate Tribunal (NCLAT) in an insolvency case after finding that the NCLT had relied on fictitious AI-generated legal citations, a lapse overlooked by the appellate tribunal. This is the latest in a series of interventions by the Court this year, cautioning against the use of AI-generated fictitious judicial precedents in court proceedings. Through its rulings and oral observations, the Court has consistently adopted a strict and cautionary approach to the deployment of AI in the justice delivery system. On February 27, the same Bench of Justices P.S. Narasimha and Alok Aradhe took cognisance of a trial court relying on AI-generated fictitious case laws and underlined that it was not merely “an error in decision-making” but amounted to judicial “misconduct”.

The Court has made clear that AI may serve as an assistive tool to improve efficiency, but it can never replace independent human reasoning, judicial discretion or professional accountability. In AI, humanity is encountering what many experts fear might well be an existential question. AI disruption is a known unknown – everyone knows that it is happening but nobody quite knows its extent or implications. There is, however, enough evidence of a combination of human stupidity and deliberate design leading to dangerous social outcomes being derived from AI. Human oversight is the essential counter to these dangers. The Court has said that presenting fabricated, machine-generated judgments to a court constitutes professional misconduct for advocates and a serious lapse of duty for judges; and any judgment influenced by an iota of fake or hallucinated AI material is “no decision in the eyes of law”. The draft ‘Regulations for Use of Artificial Intelligence (AI) in Courts, 2026’ prohibits AI in the making of judicial outcomes such as the function of adjudication, sentencing, or deciding bail eligibility and evaluating the credibility of parties or witnesses. Even as the draft is open for public consultation, the Court has directed the Bar Council of India (BCI) to set up a dedicated committee to formulate strict norms and define disciplinary actions for lawyers who cite unverified AI material. Justice must be done and seen; not hallucinated.

Manufacturing justice: Use of AI-fabricated judicial 'precedents' amounts to criminal misconduct

Supreme Court's strong warning against AI hallucinations in courts

- The Supreme Court (SC) has cautioned against the use of **AI-generated fictitious case laws, judgments, and legal citations** in judicial proceedings.

- It observed that AI hallucinations can lead to a **serious miscarriage of justice**.

The SC set aside orders of: National Company Law Tribunal and National Company Law Appellate Tribunal. In an insolvency matter, the NCLT relied on **non-existent AI-generated legal citations**.

The NCLAT failed to detect this error.

Repeated judicial concern in 2026

This is one of several interventions by the Supreme Court this year against AI-generated fake precedents. The Court has consistently adopted a **cautious approach towards AI in the justice system**.

The Court clarified that AI may Improve efficiency, Assist in research and administrative functions, However, AI **cannot replace**:, Human reasoning, Judicial discretion, Professional accountability.

Manufacturing justice: Use of AI-fabricated judicial 'precedents' amounts to criminal misconduct

Professional misconduct for lawyers

Presenting AI-generated fake judgments before a court amounts to **Professional misconduct by advocates.**

Judges relying on such material commit a **serious lapse of duty.**

The draft "**Regulations for Use of Artificial Intelligence (AI) in Courts, 2026**" prohibit AI from:

- Determining judicial outcomes.
- Adjudication of cases.
- Sentencing.
- Deciding bail eligibility.
- Assessing credibility of parties or witnesses.

Directions to the Bar Council of India

The Court directed the Bar Council of India to:

- Establish a dedicated committee.
- Frame strict norms on AI use.
- Prescribe disciplinary action against lawyers citing unverified AI-generated material.

Thank You